Dramatic Solvent Effect in the Reduction of 2,3-Allenoic Acid Esters. A Simple Synthesis of 2,3-Allenols from 2,3-Allenoates

YIN, Shao-Hu[a](尹少虎) MA, Sheng-Ming*,[b](麻生明) JIAO, Ning[b](焦宁)
TAO, Feng-Gang[a](陶凤岗)

[a] *Department of Chemistry, Fudan University, Shanghai 200433, China*

[b] *State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, Shanghai 200032, China*

2,3-Allenols were prepared conveniently from the reduction reaction of 2,3-allenoates with DIBAL-H (Diisobutylaluminum hydride) in toluene. A dramatic solvent effect was observed for this reaction.

Keywords 2,3-allenoates, reduction, 2,3-allenols

Introduction

2,3-Allenols are an important class of functionalized allenes, which can be used as the starting materials for the synthesis of oxiranes,[1] 2,5-dihydrofurans,[2] α-methylenelactone,[3] α- or γ- amino alcohols,[4] and α,β-unsaturated ketones.[5] These compounds were usually prepared by reductive elimination of mono-*O*-tetrahydropyran-2-yl or alkoxy derivatives of butyne-1,4-diols,[6] transition metal-mediated or -catalyzed reaction of 1-alkynyl epoxides,[7] the reaction of Grignard reagents with 4-hydroxy-2-alkynyl chlorides,[8] hydrolysis of 2,3-allenyl bromides,[9] treatment of allylic alcohols with $:CBr_2$ and RLi,[10] electrochemical reaction of aldehydes or ketones with propargylic bromides,[11] $CrCl_2$-mediated reaction of aldehydes or ketones with propargylic bromides,[12] and CuBr-mediated reaction of terminal propargylic alcohols with paraformaldehyde.[13] In this paper we wish to report a facile synthesis of substituted 2,3-allenols via the reduction of 2,3-allenoates.

Treatment of ethyl 2-methyl-4-phenyl-2,3-butadienoate (**1a**) with 4.0 equiv. of DIBAL-H solution in cyclohexane at room temperature afforded a complicated mixture (Entry 1, Table 1). The reaction carried out in anhydrous ether using 2 equiv. of DIBAL-H (solution in cyclohexane) at −78 ℃ afforded **2a** in 15% yield. With 3.0 equiv. of DIBAL-H in cyclohexane, the reaction in ether at −90— −140 ℃ afforded a complicated mixture (Entry 2, Table 1). However, it is interesting to note that the corresponding reaction using the DIBAL-H solution in toluene afforded **2a** cleanly in 83% yield (Entry 4, Table 1), indicating that an interesting dramatic solvent effect occurred in this reduction. Some typical results of the reduction of 2,3-allenoates with DIBAL-H in toluene are summarized in Table 2. From Table 2 it can be seen that R^1 can be aromatic, alkyl and H groups and R^2 can be aromatic, benzyl, allyl and alkyl groups and the yields range from 53% to 85%. However, with a carbon-carbon bond in the R^2 substituent, the reaction afforded 2,3-allenol **2c** in 42% yield together with an unidentified product (Entry 3, Table 2)

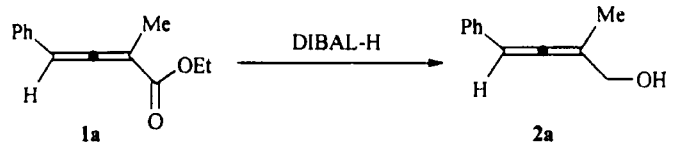
In conclusion, we have observed a dramatic solvent effect in the reduction reaction of 2,3-allenoates using DIBAL-H as the reductant. So far the reason for this is not clear yet. With the convenient synthesis of 2,3-allenoates,[14-19] this method provides a facile route to 2,3-allenols. With our recent advances in the resolution of racemic 2,3-allenoic acids, this method may provide a

* E-mail: masm@pub.sioc.ac.cn

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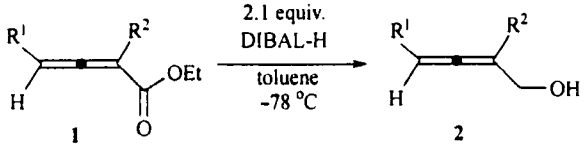
Table 1 Reduction of ethyl 2-methyl-4-phenyl-2,3-butadienoates with DIBAL-H



1a 2a

Entry	Solvent for DIBAL-H	DIBAL-H (equiv.)	Solvent	Temp. (℃)	Time (h)	Yield (%)
1	cyclohexane	4.0	-	r.t.	1.5	complicated
2	cyclohexane	2.0	ether	-78	3.0	15
3	cyclohexane	3.0	ether	-90—-140	3.5	complicated
4	toluene	2.1	toluene	-78	5.0	83

Table 2 Synthesis of substituted 2,3-allenols via the reduction of 2,3-allenoates with DIBAL-H



1 2

Entry	1			Time (h)	Yield of 2 (%)
	R^1	R^2			
1	Ph	Me	(**1a**)	5	83 (**2a**)
2	C_6H_{13}	Me	(**1b**)	3	71 (**2b**)
3	Ph	Allyl	(**1c**)	6	42 (**2c**)[a]
4	C_4H_9	Me	(**1d**)	2	76 (**2d**)
5	1-Naphthyl	Me	(**1e**)	5	57 (**2e**)
6	H	Bn	(**1f**)	3	85 (**2f**)
7	H	C_3H_7	(**1g**)	3	53 (**2g**)
8	C_3H_7	Me	(**1h**)	3	63 (**2h**)

[a] An unknown product was also formed.

facile synthesis of optically active 2,3-allenols with the axial chirality.[20]

Experimental

The starting materials 2,3-allenoates **1a**—**1h** were prepared according to the known procedure.[14-19] **1c** and **1f** are new compounds. The spectral data are listed as follows:

Ethyl 2-allyl-4-phenylbuta-2,3-dienoate (**1c**) ^{1}H NMR (300 MHz, $CDCl_3$) δ: 7.16—7.27 (m, 5H), 6.47 (t, J = 3.0 Hz, 1H), 5.75—5.84 (m, 1H), 4.95—5.10 (m, 2H), 4.09—4.17 (m, 2H), 3.03—3.08 (m, 2H), 1.17 (t, J = 6.9 Hz, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ: 212.58, 166.55, 135.01, 132.55, 129.05 (2C), 128.00, 127.53 (2C), 116.90, 103.23, 98.87, 61.46, 33.70, 14.52; IR (neat) ν: 1945, 1713, 1256 cm^{-1}; MS m/z (%): 228 (M^+, 16.47), 155 (100); HRMS calcd for $C_{15}H_{16}O_2$ 228.1150, found 228.1135.

Ethyl 2-benzylbuta-2,3-dienoate (**1f**) ^{1}H NMR (300 MHz, $CDCl_3$) δ: 7.20—7.30 (m, 5H), 5.11 (t, J = 2.5 Hz, 2H), 4.20 (q, J = 7.2 Hz, 2H), 3.59 (t, J = 2.5 Hz, 2H), 1.28 (t, J = 6.9 Hz, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ: 214.65, 167.10, 139.31, 129.15 (2C), 128.52 (2C), 126.60, 100.53, 79.49, 61.40, 35.14, 14.48; MS m/z (%): 202 (M^+, 2.27), 129 (100); IR (neat) ν: 1967, 1942, 1712, 1602, 1259 cm^{-1}; HRMS calcd for $C_{13}H_{14}O_2$ 202.0994, found 202.0996.

Synthesis of 2-methyl-4-phenylbuta-2,3-dienol (**2a**)

Typical procedure

DIBAL-H (2.1 mL, 1.0 mol/L solution in toluene, 2.1 mmol) was added dropwise to a solution of **1a** (202 mg, 1.0 mmol) in dry toluene (1 mL). After the reaction was complete as monitored by TLC (eluent: hexane/ethyl acetate = 6:1, V:V), the mixture was quenched with 0.5 mL of methanol, washed with 25 mL of aqueous HCl solution (5%). The organic layer was separated and the aqueous layer was extracted by ether. The combined organic layer was dried over Na_2SO_4, filtered and evaporated. The residue was purified by column chromatography on silica gel (eluent: hexane/ethyl acetate = 6:1, V:V) to give 133 mg (83%) of **2a**:[7a] white solid, m.p.

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57—58 ℃(petroleum ether); ^{1}H NMR (300 MHz, $CDCl_3$) δ: 7.21—7.32 (m, 5H), 6.27 (q, J = 2.7 Hz, 1H), 4.14—4.17 (m, 2H), 2.01 (s, 1H), 1.85 (d, J = 2.7 Hz, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ: 201.07, 134.98, 128.86 (2C), 127.29, 127.02 (2C), 104.95, 97.27, 64.09, 15.69; IR (KBr) ν: 3244, 1951 cm^{-1}; MS m/z (%): 161 (M^+ + 1, 4.36), 160 (M^+, 33.18), 128 (100).

2-*Methyldeca*-2,3-*dien*-1-*ol* (**2b**)

Starting from **1b** (210 mg, 1.0 mmol) in toluene (1 mL) and DIBAL-H in toluene (2.1 mL, 1.0 mol/L, 2.1 mmol) to afford 119 mg (71%) of **2b**: yellow oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ: 5.25—5.28 (m, 1H), 3.98 (s, 2H), 2.00(q, J = 6.7 Hz, 2H), 1.69 (d, J = 2.7 Hz, 3H), 1.50 (brs, 1H), 1.28—1.40 (m, 8H), 0.88 (t, J = 6.6 Hz, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ: 199.25, 100.17, 93.96, 63.85, 31.60, 29.06, 29.04, 28.66, 22.57, 15.56, 14.00; IR (neat) ν: 3333, 2927, 1968 cm^{-1}; MS m/z (%): 168 (M^+, 3), 43 (100); HRMS calcd for $C_{11}H_{20}O$ 168.1514, found 168.1534.

2-*Allyl*-4-*phenylbuta*-2,3-*dien*-1-*ol* (**2c**)

Starting from **1c** (228 mg, 1.0 mmol) in toluene (1 mL) and DIBAL-H in toluene (2.1 mL, 1.0 mol/L, 2.1 mmol) to afford 75 mg (42 %) of **2c**: yellow oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ: 7.22—7.34 (m, 5H), 6.35—6.37 (m, 1H), 5.87—5.96 (m, 1H), 5.20 (d, J = 17.0 Hz, 1H), 5.10 (d, J = 9.0 Hz, 1H), 4.19—4.22 (m, 2H), 2.95 (d, J = 6.6 Hz, 2H), 1.95 (brs, 1H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ: 201.52, 135.32, 134.82, 128.92 (2C), 127.39 (2C), 126.99, 116.93, 108.17, 98.24, 62.96, 34.55; IR (neat) ν: 3352, 2926, 1952 cm^{-1}; MS m/z (%): 186 (M^+, 31.68), 187 (M^+ + 1, 4.67), 155 (100); HRMS calcd for $C_{13}H_{14}O$ 186.1044, found 186.1013.

2-*Methylocta*-2,3-*dien*-1-*ol* (**2d**)[7b]

Starting from **1d** (151 mg, 0.83 mmol) and DIBAL in toluene (2.1 mL, 1.0 mol/L, 2.1 mmol) to afford 88 mg (76%) of **2d**: yellow oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ: 5.26—5.28 (m, 1H), 3.98 (s, 2H), 2.00 (q, J = 6.9 Hz, 2H), 1.69 (d, J = 2.7 Hz, 3H), 1.50 (brs, 1H), 1.32—1.39 (m, 4H), 0.89 (t, J = 5.4 Hz, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ: 199.09, 100.35, 94.21, 63.81, 31.26, 28.77, 22.07, 15.70, 13.85; IR (neat) ν: 3332, 2928, 1968, 1012 cm^{-1}; MS m/z (%): 140 (M^+, 0.34), 98 (100), 107 (11.61), 83 (77.21).

2-*Methyl*-4-(1′-*naphthyl*)*buta*-2,3-*dien*-1-*ol* (**2e**)

Starting from **1e** (133 mg, 0.5 mmol) and DIBAL in toluene (1.08 mL, 1.0 mol/L, 1.08 mmol) to afford 65 mg (57%) of **2e**: white solid, m.p.: 56—57 ℃ (petroleum ether); ^{1}H NMR (300 MHz, $CDCl_3$) δ: 8.23 (d, J = 9.0 Hz, 1H), 7.87 (d, J = 9.6 Hz, 1H), 7.75 (d, J = 8.4 Hz, 1H), 7.42—7.57 (m, 4H), 6.96—6.98 (m, 1H), 4.20—4.22 (m, 2H), 1.93 (d, J = 2.7 Hz, 3H), 1.85 (brs, 1H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ: 202.75, 134.14, 131.22, 131.04, 128.93, 127.87, 126.35, 125.98, 125.84, 125.72, 123.78, 103.65, 93.82, 64.23 15.80; IR (KBr) ν: 3354, 2923, 1954, 1013, 790, 774 cm^{-1}; MS m/z (%): 210 (M^+, 17.88), 192 (100), 178 (93.45), 152 (60.64); HRMS calcd for $C_{15}H_{14}O$ 210.1045, found 210.1015.

2-*Benzylbuta*-2,3-*dien*-1-*ol* (**2f**)[21]

Starting from **1f** (5.05 g, 25 mmol) in toluene (70 mL) and DIBAL-H in toluene (52.5 mL, 1.0 mol/L, 52.5 mmol) to afford 3.42 g (85%) of **2f**: yellow oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ: 7.05—7.50 (m, 5H), 4.65—5.05 (m, 2H), 4.03 (s, 2H), 3.39 (s, 2H), 1.70 (brs, 1H); IR (neat) ν: 3347, 2912, 1951, 1599, 1491 cm^{-1}; MS m/z (%): 160 (M^+, 59), 115 (100).

2-*Propylbuta*-2,3-*dien*-1-*ol* (**2g**)[7b]

Starting from **1g** (7.70 g, 50 mmol) in toluene (150 mL) and DIBAL in toluene (2.1 equiv.) to afford 2.94 g (53%) of **2g**: yellow oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ: 4.71—4.98 (m, 2H), 4.02—4.04 (m, 2H), 1.82—2.09 (m, 2H), 1.57 (brs, 1H), 1.32—1.55 (m, 2H), 0.98 (t, J = 7.3 Hz, 3H); IR (neat) ν: 3387, 2962, 1958 cm^{-1}; MS m/z (%): 112 (M^+, 4), 95 (100).

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2-Methylhepta-2,3-dien-1-ol (**2h**)[8,9]

Starting from **1h** (2.60 g, 16.8 mmol) to afford 1.34 g (63 %) of **2h**: yellow oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ: 5.19—5.31 (m, 1H), 3.98 (s, 2H), 1.98 (q, J = 7.08 Hz, 2H), 1.70 (d, J = 2.9 Hz, 3H), 1.56 (brs, 1H), 1.31—1.54 (m, 2H), 0.92 (t, J = 7.3 Hz, 3H); IR (neat) ν: 3317, 2959, 1967 cm^{-1}; MS m/z (%): 126 (M^+, 7), 84 (100).

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